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Sensorless displacement estimation for an electrostatic film motor using driving currents

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Abstract. This work proposes a self-sensing method for a synchronous electrostatic film motor. The method utilizes the relation between the driving voltages and currents of the motor. Through several signal conversions on the measured motor currents, the system distills the information of the slider position to realize displacement sensing. The resulting system can track the slider position both for synchronous and non-synchronous operations. The system assumes stepwise motions of the slider and fully exploits the characteristic current features of the motions. Two different signal conversion methods have been tested using numerically simulated motor currents and real motor currents. The two methods differ in terms of accuracy and noise immunity. Considering that the currents of the electrostatic motor are relatively small, the method with high noise immunity and relatively low accuracy was found useful for practical applications. The position resolution achieved by the method was in the order of the electrode pitch, which is 200 μm in the prototype motor used in this work.

Keywords: Electrostatic film motor, displacement estimation, sensorless

1. Introduction

Electrostatic film motors share the same underlying principles with electrostatic micro-motors [1–3] that have been developed over decades. Their different size ranges, however, make a huge difference in their applications. While the applications of micro-motors can be found in MEMS systems and micro-robot systems, the electrostatic film motors can be used for typical mechatronic devices of ordinary size. The motor are typically fabricated using thin films or thin boards that measure from several centimeters up to tens of centimeters in size, on which fine-patterned electrodes are fabricated. Such large sizes allow them to produce much larger forces than micro-motors; the output force can easily reach several newtons, or even more, in linear-motion types. Various operation principles have been proposed for the film motors [4–7], and the most typical one among them is a synchronous principle called *dual excitation multiphase electrostatic drive* [8,9].

The output performance of the synchronous motor depends on the applied voltage; higher voltages result in larger outputs. The maximum applicable voltages greatly depend on the surrounding environment. If operated in the atmosphere, discharges in the air limit the maximum voltages, and in turn, the output performance. However, if operated in dielectric liquid that allows applications of higher voltages, the

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motors can exhibit superior performance, which is even comparable to conventional electromagnetic linear motors. Especially, by stacking many pairs of films, the total output of the motor can be considerably enhanced, which is suitable for power applications such as artificial muscles of robots [10] or positioning stages in mechatronic systems.

Regardless of the surrounding environments, the film motors can possess unique features, such as thin structure, optical transparency, and/or mechanical flexibility. Utilizing those unique features, the motors are expected to be utilized for, e.g., computer-human interaction (CHI) systems [7,11] or wall climbing robots [12]. In those applications, to fully exploit the unique features rather than the output performance, the motor is utilized in the simplest configuration; on the other hand, the required force or positioning accuracy are not very high. Those applications require displacement sensors that do not spoil the unique features of the motors. The sensors should also be thin, transparent, and/or flexible, which cannot be realized with off-the-shelf displacement sensors. Therefore, in this paper, to meet these requirements, we propose a self-sensing method, in which the slider motion is estimated from the driving currents flowing through the motor.

In previous studies, built-in sensor systems for the electrostatic film motors have already been studied. In one type, exclusive sensor electrodes are fabricated on the same films as the actuator electrodes [13,14], which work as capacitive linear encoders. Although this can realize sensing without spoiling the motor's unique features, it reduces the area of the actuator electrodes and weakens output performance per film. On the other hand, in [15], a built-in sensing system that directly utilizes actuator electrodes was proposed. In the system, a high-frequency sensing signal is superposed on low-frequency and high-voltage actuating signals, and the slider position is estimated by distilling and processing the high-frequency signal. Although this method can realize robust and precise sensing without deteriorating the actuator performance, the overall system tends to be complicated due to the requirements of high frequency signal source and transformers for signal superposition and detection.

On the other hand, if we look at electromagnetic motors, many kinds of sensorless displacement sensing methods have been studied, which are based on the driving voltages and resulting currents [16–20]. Based on a similar concept, we previously proposed a new displacement sensing system for the synchronous electrostatic film motor that directly utilizes the driving currents of the motor [21]. Different from electromagnetic motors, the driving currents of the electrostatic motor are small, which makes it difficult to realize high sensing accuracy. Resulting accuracy in the previous study was equivalent to the electrode pitch, which was worse than the previous studies about built-in sensing of the electrostatic film motors. However, some of the applications mentioned above do not require very high sensing accuracy, and therefore, the new simple system can be useful for those applications.

The sensing system in the previous work, however, could detect the slider position only during synchronous operations. Considering the nature of synchronous motors, the position during synchronous motion can be easily known even without sensors. Therefore, in that sense, the developed system was not really useful in practical applications. For practical applications, the sensor should also be able to track the slider position during "out-of-synchronous" operations. Therefore, in this paper, we propose a new signal processing scheme to detect the slider position regardless of operation status. The target motor is the synchronous electrostatic film motor. Although the synchronous motor has a variant that utilizes skewed electrode configuration to realize smooth motion [22], this work focuses on a regular motor that utilized non-skewed parallel electrodes. Those regular motors typically create stepwise motions, and the proposed method will exploit the characteristic currents of them.

The rest of the paper is structured as follows. In the next section, the principle of the synchronous electrostatic film motor is introduced with its mathematical model. Section 3 analyzes the driving currents

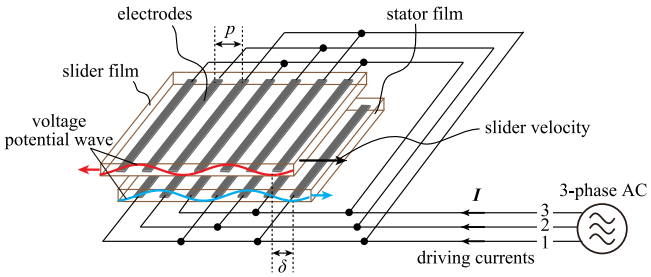


Fig. 1. Basic structure and driving principle of electrostatic film motor.

during synchronous operation to extract the relationship between the driving currents and the slider position based on Clarke and Park transformations, and proposes two methods to realize displacement estimation. The proposed methods are simulated in Section 4, to demonstrate their accuracy and noise margin. Section 5 verifies the proposed methods by using the real motor currents. Section 6 concludes this work.

2. Electrostatic film motor

2.1. Basic structure

The basic structure of the electrostatic film motor, in the linear configuration, is shown in Fig. 1. It consists of two flexible circuit films, which work as a stator and a slider. In most cases, the two films are simply stacked with some minute glass beads between them for lubrication. In each film, three-phase electrodes are arranged in parallel with the same regular pitch, p, which was typically around 200 μm in most of the past studies. When a 3-phase AC voltage, with an amplitude of around 1 to 2 kV, is applied to both of those 3-phase electrodes as in Fig. 1, two voltage potential waves are excited in the films, which move toward different directions. The two potential waves create an electrostatic attractive force between the two films, whose tangential component propels the slider. The electrostatic force depends on the spatial phase difference of the two waves, which is denoted by δ in the rest of the paper.

2.2. Capacitance coefficients matrix

As the motor is operated by a 3-phase voltage, the motor can be modeled as a three-terminal capacitance network [22]. The capacitance network can be mathematically represented by a symmetric matrix called *capacitance coefficients matrix*, **C**. The elements in the matrix depend on the motor structure, and they are functions of the slider position. Ignoring higher harmonics, the elements can be expressed as [21]

$$\begin{cases} C_{11} = C_{st0} + C_{sl0} + 2C_m(\theta_x - 2\pi/3) \\ C_{22} = C_{st0} + C_{sl0} + 2C_m(\theta_x) \\ C_{33} = C_{st0} + C_{sl0} + 2C_m(\theta_x + 2\pi/3) \\ C_{12} = -C_{t0} - C_{l0} + 2C_m(\theta_x + 2\pi/3) \\ C_{13} = -C_{t0} - C_{l0} + 2C_m(\theta_x) \\ C_{23} = -C_{t0} - C_{l0} + 2C_m(\theta_x - 2\pi/3) \end{cases} \tag{1}$$

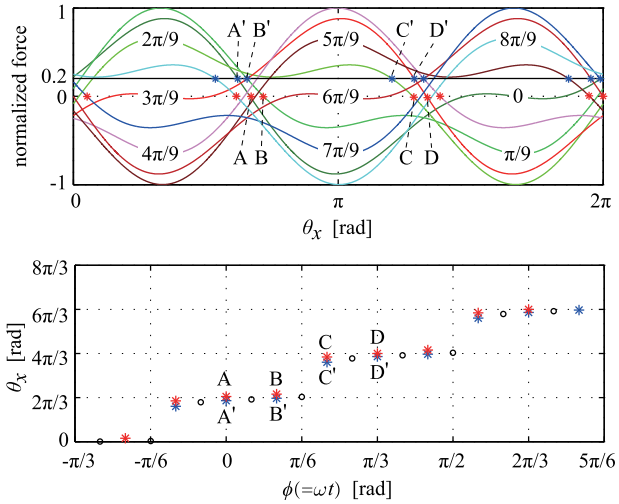


Fig. 2. Normalized theoretical force at different voltage phases (upper) and the corresponding slider motion (lower). In the upper plot, the force curves that correspond to the circle markers in the lower plot were omitted for visibility.

where C_{st0}, C_{sl0}, C_{t0}, C_{l0}, C_{m0}, C_{m1} and C_{m2} are all positive constants, and $C_m(\theta_x) = -C_{m0} - C_{m1}\cos(\theta_x) - C_{m2}\cos(2\theta_x)$. Here, θ_x expresses the slider position x in its electric angle representation, that is $\theta_x = 2\pi x/3p$.

From the electrostatic coenergy of the capacitive system, the thrust force of the motor can be calculated as

$$f_{thrust} = \frac{1}{2}\mathbf{V}^T\frac{d\mathbf{C}}{dx}\mathbf{V} = \frac{3\pi}{2p}V_0^2(C_{m1}\sin\delta + 2C_{m2}\sin(2\theta_x + 2\omega t - \pi/3)) \tag{2}$$

where δ is the spatial phase difference defined as

$$\delta = \theta_x - 2\omega t + \pi/3 \tag{3}$$

and $\mathbf{V}$ is the applied voltage, which is expressed as

$$\mathbf{V} = V_0[\sin(\omega t), \sin(\omega t - 2\pi/3), \sin(\omega t + 2\pi/3)]^T \tag{4}$$

with V_0 being the amplitude and ω being the angular frequency of the AC voltage.

Figure 2 shows the force characteristics for different voltage phases, while the amplitude, V_0, is fixed. The capacitances used for the calculation are adopted from the previous study [22]. The slider motion can be explained using this figure. If we assume quasi-static operation, the slider position can be found at the stable equilibrium of the force curve for each voltage phase. In no-load condition (force = 0), for example, the slider position will be at point A when the phase $\phi = \omega t$ is zero, and then it shifts to the points B, C, and D as the phase increases. Since there is a large gap between the points B and C, the resulting slider motion will be stepwise, as shown in the lower plot of Fig. 2. In a loaded condition, the equilibrium points will shift to A', B', C', and so on, where the load matches with the thrust force. The lower plot shows the slider motion analyzed as explained above. By plotting the equilibrium points against the phase, which can be regarded as time, we can know the time-course trajectory of the slider. As shown in this plot, the slider motion is stepwise regardless of the load. Here, the average speed of the step motions is called "synchronous speed" and is $6pf = 3p\omega/\pi$, where f is the frequency of the applied AC. If the load exceeds

the maximum of a thrust force curve, the equilibrium does not exist anymore, and the slider will step out from the synchronous motion, which is called "out-of-step" or "out-of-synchronization".

3. Current analysis

3.1. Driving currents

In this section, we derive the driving currents flowing through the motor and analyze them for distilling the displacement information. For simplicity of calculation, we assume $C_{m2} = 0$ in the following.

The capacitance matrix can be divided into two parts

$$\mathbf{C} = \mathbf{C}_0 + \mathbf{C}_x \tag{5}$$

where

$$\mathbf{C}_0 = \begin{cases} C_{11}(= C_{22} = C_{33}) = C_{st0} + C_{sl0} - 2C_{m0} \\ C_{12}(= C_{13} = C_{23}) = -C_{t0} - C_{l0} - 2C_{m0} \end{cases}$$

is a constant matrix, and

$$\mathbf{C}_x = \begin{cases} C_{11}(= C_{23}) = -2C_{m1}\cos(\theta_x - 2\pi/3) \\ C_{22}(= C_{13}) = -2C_{m1}\cos(\theta_x) \\ C_{33}(= C_{12}) = -2C_{m1}\cos(\theta_x + 2\pi/3) \end{cases}$$

is a function of the slider position, θ_x.

As the electric charges, $\mathbf{Q}$, on the three terminals of the motor can be calculated using the matrix as

$$\mathbf{Q} = \mathbf{C}\mathbf{V} \tag{6}$$

the currents, $\mathbf{I}$, are obtained as the derivative of $\mathbf{Q}$, that is

$$\mathbf{I} = \frac{d\mathbf{Q}}{dt}. \tag{7}$$

Using Eq. (5), the currents can be also divided into two components

$$\mathbf{I} = \frac{d((\mathbf{C}_0 + \mathbf{C}_x)\mathbf{V})}{dt} = \mathbf{I}_1 + \mathbf{I}_2 \rightarrow \begin{cases} \mathbf{I}_1 = \mathbf{C}_0 \dfrac{d\mathbf{V}}{dt} \\ \mathbf{I}_2 = \dfrac{d(\mathbf{C}_x\mathbf{V})}{dt}. \end{cases} \tag{8}$$

Here, $\mathbf{I}_1$ is a pure 3-phase sinusoidal with a constant amplitude, which is here-in-after called "base current". The base current, $\mathbf{I}_1$, is calculated as

$$\mathbf{I}_1 = \begin{bmatrix} i_{u1} \\ i_{v1} \\ i_{w1} \end{bmatrix} = A \begin{bmatrix} \cos(\omega t) \\ \cos(\omega t - 2\pi/3) \\ \cos(\omega t + 2\pi/3) \end{bmatrix} \tag{9}$$

where, $A = (C_{st0} + C_{t0} + C_{sl0} + C_{l0})V_0\omega$.

The other component, $\mathbf{I}_2$, is associated with the slider motion, and is called "feature current", in the following. Since the matrix $\mathbf{C}_x$ is a function of the slider position θ_x that changes with time, the feature

current, $\mathbf{I}_2$, can be further deformed as

$$\mathbf{I}_2 = \frac{d\mathbf{C}_x\mathbf{V}}{dt} = \omega_x \frac{d\mathbf{C}_x}{d\theta_x}\mathbf{V} + \mathbf{C}_x\frac{d\mathbf{V}}{dt} \tag{10}$$

where $\omega_x = d\theta_x/dt$ is the slider velocity. Substituting Eq. (4) to Eq. (10), $\mathbf{I}_2$ can be expressed as

$$\mathbf{I}_2 = \begin{bmatrix} i_{u2} \\ i_{v2} \\ i_{w2} \end{bmatrix} = -B \begin{bmatrix} \cos(\theta_x - \omega t + \pi/3) \\ \cos(\theta_x - \omega t - \pi/3) \\ \cos(\theta_x - \omega t + \pi) \end{bmatrix} \tag{11}$$

where $B = 3C_{m1}\ V_0(\omega_x - \omega)$.

Then, the total currents, $\mathbf{I}$, in Eq. (8) can be expressed as

$$\mathbf{I} = \begin{bmatrix} i_u \\ i_v \\ i_w \end{bmatrix} = \begin{bmatrix} A\cos(\omega t) - B\cos(\theta_x - \omega t + \pi/3) \\ A\cos(\omega t - 2\pi/3) - B\cos(\theta_x - \omega t - \pi/3) \\ A\cos(\omega t + 2\pi/3) - B\cos(\theta_x - \omega t + \pi) \end{bmatrix}. \tag{12}$$

As this 3-phase expression is redundant, as well as it is complicated as it includes ωt, the following subsection applies Clarke and Park transformations that will lead to an easy extraction of the position θ_x.

3.2. Signal conversion for displacement sensing

In the above, the 3-phase current of the electrostatic motor was expressed in a 3-phase coordinate system (u, v, w) as $[i_u, i_v, i_w]^T$. Similarly to the current analysis of the 3-phase AC magnetic motors, this work applies Clarke and Park transformations to simplify the expression. First, Clarke transformation is applied to transform the 3-phase current vector into a 2-phase vector defined on two orthogonal axes, (α, β), using the following transformation [23]

$$\begin{cases} i_\alpha = i_u \\ i_\beta = \frac{1}{\sqrt{3}} i_u + \frac{2}{\sqrt{3}} i_v. \end{cases} \tag{13}$$

With Eqs (12) and (13), the 2-phase expression in the (α, β) frame is obtained as

$$\mathbf{I}_{\alpha\beta} = \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \begin{bmatrix} A\cos(\omega t) - B\cos(\theta_x - \omega t + \pi/3) \\ A\sin(\omega t) - B\sin(\theta_x - \omega t + \pi/3) \end{bmatrix} = \begin{bmatrix} A\cos(\omega t) - B\cos(\omega t + \delta) \\ A\sin(\omega t) - B\sin(\omega t + \delta) \end{bmatrix}. \tag{14}$$

Next, in order to remove ωt in Eq. (14), Park transformation is utilized that maps the vector from the (α, β) frame to a rotating reference frame (d, q). The rotating matrix from (α, β) to (d, q) is [23]

$$\mathbf{R} = \begin{bmatrix} \cos(\omega t) & \sin(\omega t) \\ -\sin(\omega t) & \cos(\omega t) \end{bmatrix}. \tag{15}$$

The expression of the driving currents in the (d, q) frame is obtained as

$$\mathbf{I}_{dq} = \begin{bmatrix} i_d \\ i_q \end{bmatrix} = \mathbf{R}\mathbf{I}_{\alpha\beta} = \begin{bmatrix} A - B\cos\delta \\ -B\sin\delta \end{bmatrix}. \tag{16}$$

In this expression, the spatial phase difference δ can be easily derived as

$$\delta = \arctan\left(\frac{i_q}{i_d - A}\right). \tag{17}$$

Using δ, the slider position is obtained as

$$x = \frac{3p}{2\pi}(2\omega t + \delta - \pi/3). \tag{18}$$

In actual implementations of Eq. (17), the range of arctan needs to be carefully considered. Typically, arctan is calculated in a single complete range of $(-\pi/2, \pi/2)$. On the other hand, δ can take any value within $(-\infty, \infty)$. Therefore in calculating δ, the result of arctan needs to be phase-unwrapped using an unwrapping algorithm such as in [24,25].

3.3. Modified signal conversion

To calculate the slider position based on Eq. (17), we need to correctly identify A, which is the amplitude of the base current. This is, however, not practical, as the correct identification of A is considerably difficult due to its small size. In addition, it will be easily affected by noise in the measured currents. Therefore, we propose to eliminate the base current electrically by using a high pass filter (HPF) before the calculation.

From Eq. (9), the angular frequency of the base current is found as ω, which is the frequency of the applied voltage. For the feature current, the angular frequency is found from Eq. (11) as $\omega_x - \omega$, as $\theta_x = \omega_x t$. If we assume stepwise motion as in Fig. 2, the velocity of the slider changes between two values, nearly zero for the plateau of the steps, and a considerably large value (which is approximated as ∞ in the following) for the jumping between two steps. Therefore, the angular frequency of the feature current changes between ω and ∞.

Here, we assume that we apply HPF with the cutoff angular frequency slightly higher than (or almost equal to) ω. Then, the filter affects both the base current and the feature current, and the resulting total current, $\mathbf{I}_f$ can be expressed as

$$\mathbf{I}_f = n\mathbf{I}_1^{\angle\alpha} + m\mathbf{I}_2^{\angle\beta} \tag{19}$$

where n and m express the filter gains at their frequencies and are within the range (0,1), and superscripts $\angle\alpha$ and $\angle\beta$ mean that the phases are delayed by α and β by HPF.

After Clarke and Park transformations based on Eqs (14) and (16), the filtered current in Eq. (19) becomes

$$\mathbf{I}_{dqf} = \begin{bmatrix} i_{df} \\ i_{qf} \end{bmatrix} = \begin{bmatrix} nA\cos\alpha - mB\cos(\delta - \beta) \\ -nA\sin\alpha - mB\sin(\delta - \beta) \end{bmatrix}. \tag{20}$$

By taking arctan of the two components, an estimation of the spatial phase difference $\tilde{\delta}$ is obtained.

$$\tilde{\delta} = \arctan\left(\frac{i_{qf}}{i_{df}}\right) = \arctan\left(\frac{-nA\sin\alpha - mB\sin(\delta - \beta)}{nA\cos\alpha - mB\cos(\delta - \beta)}\right). \tag{21}$$

As explained above, the frequency of the feature current depends on the slider velocity, and the slider velocity changes between two different velocities. Therefore, the behavior of $\tilde{\delta}$ should be discussed for the two velocities.

[**Case 1:** ($\omega_x \approx 0$)] In this case, the frequencies of the base current and feature current are the same; therefore, $n = m$ and $\alpha = \beta$. From [22], it is known that the capacitance coefficients in A and B have a relationship of $(C_{st0} + C_{t0} + C_{sl0} + C_{l0}) \approx 20C_{m1}$. Therefore, we have $nA \gg mB$, which leads to $\tilde{\delta} \approx -\alpha$.

[**Case 2:** ($\omega_x \approx \infty$)] As B contains ω_x, we have $B \to \infty$, and therefore $|nA| \ll |mB|$. In addition, as the feature current frequency becomes considerably high, the phase delay caused by HPF will nearly be zero, that is $\beta \approx 0$. In this case, Eq. (21) gives $\tilde{\delta} \approx \delta$.

Then, using $\tilde{\delta}$, the slider position is estimated as

$$\tilde{x} = \frac{3p}{2\pi}(2\omega t + \tilde{\delta} - \pi/3) \tag{22}$$

which will be the output of the sensing system.

The error of the estimated position ($\epsilon_f = \tilde{x} - x$) is proportional to the estimation error of the spatial phase difference, $\tilde{\delta} - \delta$. For case 1, if we assume that the initial position is zero ($\theta_x = 0$), the error is calculated as

$$\tilde{\delta} - \delta = -\alpha + 2\omega t - \pi/3. \tag{23}$$

The equation suggests that the error grows with time. Therefore, if this situation, which is $\omega_x \approx 0$, lasts for a long time, the error will explode and the displacement estimation will completely fail. In many practical cases, however, the slider moves as the phase changes for $\pi/3$, according to Fig. 2, and thus the error is bounded in the range $(-\alpha - \pi/3, -\alpha + \pi/3)$. As the error is nearly zero for case 2, the overall error in $\tilde{\delta}$ is in the range $(-\alpha - \pi/3, \max(0, -\alpha + \pi/3))$. If we assume a second-order filter for the HPF, the phase delay α is smaller than π, which gives the maximum error range of $(-4\pi/3, 0)$. This error range corresponds to the position error of $(-2p, 0)$. By adding an offset of p, the resulting error is $(-p, p)$, or plus/minus one electrode pitch, which falls into the required sensor resolution.

It should be noted, however, the above discussion is not valid if the slider is held stationary by some external force. In such a case, the error shown in Eq. (23) will continuously grow and the calculation will not provide correct estimation anymore. Therefore, applications must be carefully examined when we use this modified method.

4. Simulation

4.1. Block diagram

Two different signal processing methods have been proposed in the previous section. This section verifies the methods through numerical simulations. Hereinafter, the first method without filtering is called "method 1", which is based on Eqs (17) and (18), and the modified version based on Eqs (21) and (22) is called "method 2".

Numerical simulations were performed on Simulink/Matlab 7.1. The block diagram of the simulation is shown in Fig. 3. The area enclosed by the green dashed line represents the electrostatic motor. In these blocks, with a given voltage vector, the thrust force is calculated based on Eq. (2), which is then used to simulate the motion. The two areas enclosed by the red dashed lines are the two signal processing methods for displacement estimation. The motor driving currents simulated using Eqs (6) and (7) are fed to the two methods. To check the immunity against noise, noise was added to the currents in some sets of simulations.

To check the sensor system behavior in out-of-steps, an elastic obstacle was assumed that will block the motor motion as a wall. The force from the obstacle is

$$f_{disturb} = \begin{cases} k(x - x_{obstacle}) & \text{when } x > x_{obstacle} \\ 0 & \text{otherwise} \end{cases} \tag{24}$$

where k expresses the spring coefficient of the elastic object, x is the slider position, and $x_{obstacle}$ denotes the wall position.

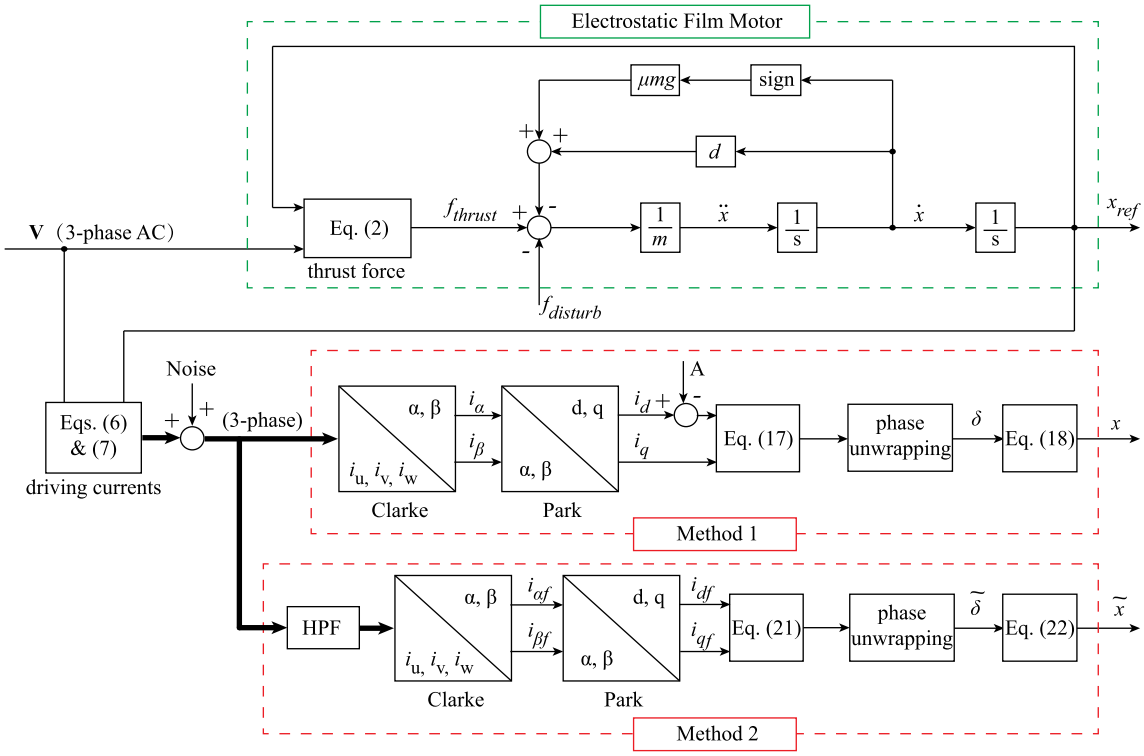


Fig. 3. The block diagram of simulation, including the electrostatic film motor and the proposed sensorless displacement estimation methods.

The capacitances used for the simulations are adopted from [22], which are

$$\begin{cases} C_{st0} = 540,\ C_{sl0} = 314,\ C_{t0} = 180,\ C_{l0} = 67, \\ C_{m0} = 60,\ C_{m1} = 50,\ C_{m2} = 15 \end{cases} \tag{25}$$

in units of [pF]. The electrode pitch was $p = 0.2$ mm, the slider mass was $m = 6$ g, the viscous damping of the dielectric fluid was $d = 30$ N/(m/s), and the friction index was $\mu = 0.3$. The obstacle was placed at $x_{obstacle} = 1.1$ mm and its spring coefficient was $k = 10$ N/mm. The 3-phase AC voltage had an amplitude $V_0 = 1500$ V and the frequency was 5 Hz. All the simulations were done using a fixed-step 5th-order Dormand-Prince method, with a time step of 0.1 ms.

4.2. *Results for method 1*

The reference slider position x_{ref}, which is used as the ground-truth, and the calculated driving currents are shown in Fig. 4. The significant spike currents come from Eq. (10) due to the stepwise movement of the motor. The motor operated synchronously until around 0.2 s, then hit the wall, and repeated out-of-step motions afterwards.

Figure 5 shows the results of method 1 without noise. After Clarke and Park transformations, the driving currents were transferred to $\mathbf{I}_{dq}$, as shown in Fig. 5(a). After removing A as in Fig. 5(b), δ was calculated according to Eq. (17), as shown in Fig. 5(c). Since δ was wrapped within $(-\pi/2, \pi/2)$, it was unwrapped to eliminate discontinuity, which gives Fig. 5(d). From Fig. 5(d), it is known that the spatial phase difference

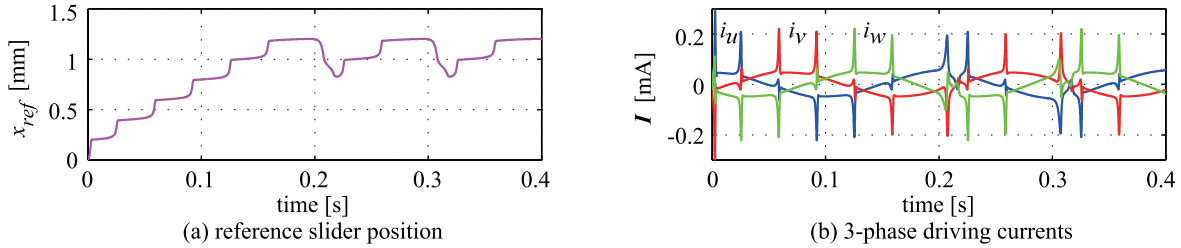


Fig. 4. The simulated slider position and the driving currents.

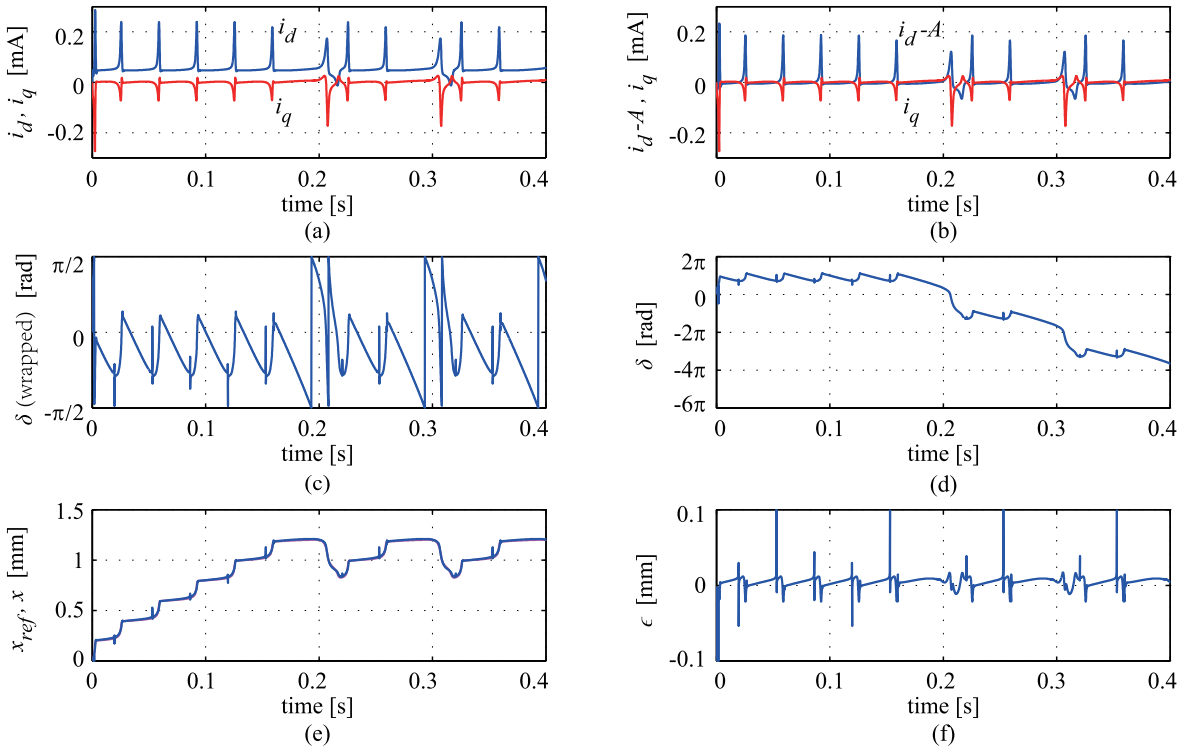


Fig. 5. The signal processing based on method 1 (noise = 0).

δ is kept almost constant during synchronous operation, and shifts as the slider steps-out. Finally, the slider position was calculated based on Eq. (18), as shown in Fig. 5(e) using a blue line. For comparison, the reference position, x_{ref} was also plotted in Fig. 5(e) using a purple line, but not distinguishable as it almost coincides with the blue line. The error $\epsilon = x_{ref} - x$ is plotted in Fig. 5(f). Although small spikes are sometimes observed in the error plot, method 1 could accurately estimate the slider position.

Next, simulations for noisy environment were carried out. As the noise source, we used the band-limited white noise block in Simulink, whose sampling time was the same with the system. For evaluation, the signal-to-noise ratio (SNR) was calculated as SNR = $20\log_{10}(A_{signal}/A_{noise})$ with a unit of dB, where A_{signal} and A_{noise} are root mean square (RMS) amplitudes of the signal and noise.

Simulations were done with different SNRs, whose interval was 1 dB. When SNR $\geq$ 59 dB, method 1 works well, as shown in Fig. 6(a). However, when SNR $<$ 59 dB, large errors were observed. In the example

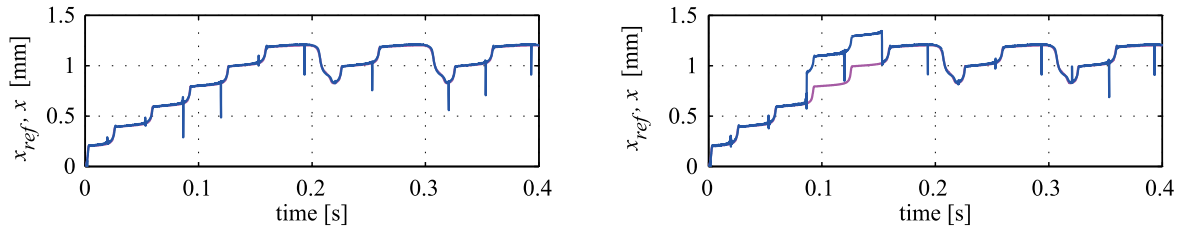


Fig. 6. Estimated position based on method 1, when noise was added to driving currents (SNR = 59 dB and SNR = 54 dB).

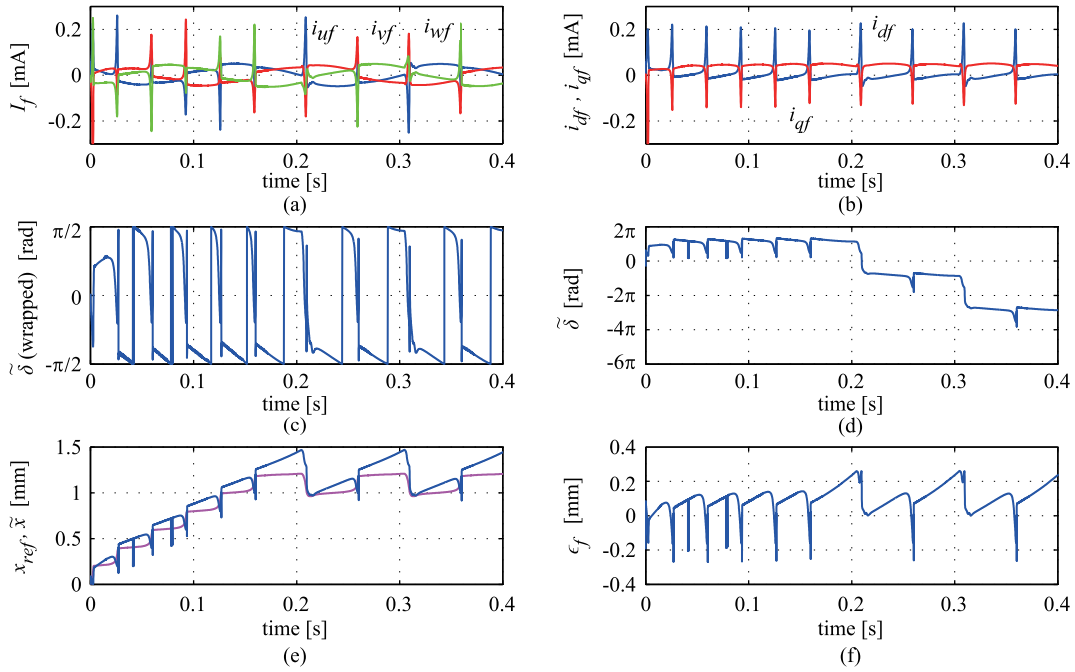


Fig. 7. The signal processing based on method 2 (noise = 0).

shown in Fig. 6(b), the estimated position still reflects the trend of the reference position; however, the output occasionally jumps. This is because $i_d - A$ in Eq. (17) was too small and was thus be drastically affected by noise. These results indicate that noise immunity is not high in method 1.

4.3. Results for method 2

A result for method 2 without noise is shown in Fig. 7. The HPF used was a second-order Butterworth filter, with a cutoff frequency of 5 Hz. Figure 7(e) shows that the estimated position almost linearly increased when the slider was resting at a constant position. This corresponds to case 1 discussed in the above section. However, when the slider jumped to move to the next step, the error ϵ_f in the estimated position was rewound. The error ϵ_f became large at around 0.2 s and 0.3 s, which is because the slider position did not change for more than $\pi/3$ of the voltage phase, and in turn, the duration of case 1 was extended. However, the increased error ϵ_f was resolved when the slider stepped out of synchronization

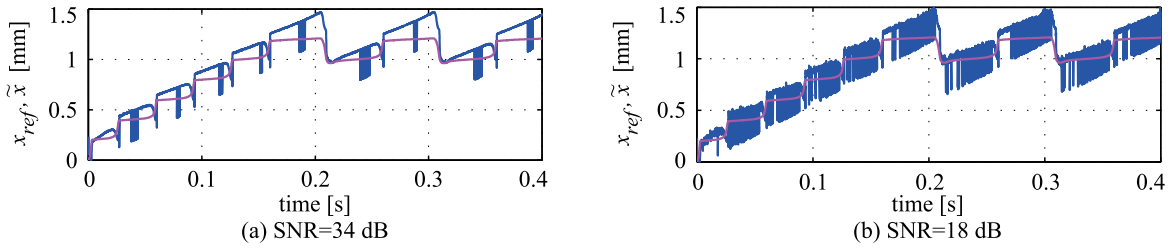


Fig. 8. Estimated position based on method 2, when noise was added to driving currents (SNR = 34 dB and SNR = 18 dB).

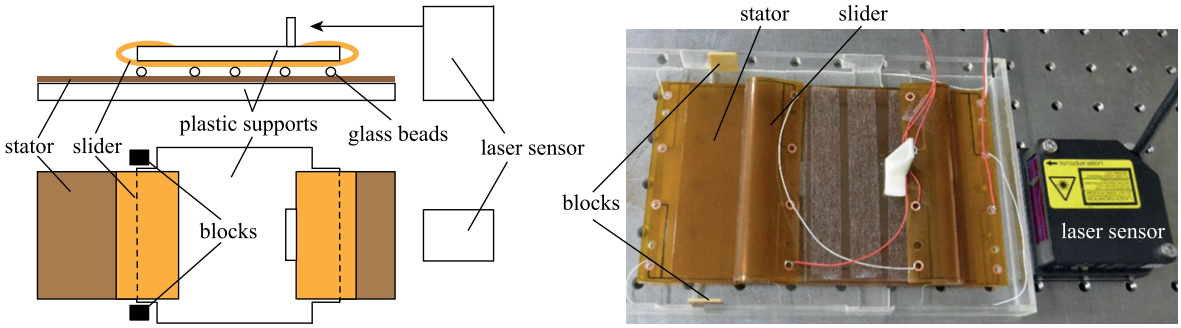


Fig. 9. Schematic diagram and photograph of experiment apparatus.

and jumped back. Except for these particular points, the error ϵ_f was almost bound by $\pm p$, which verifies the theoretical discussions.

In simulation with noise, the method 2 was found robust against noise; it could track the position when SNR was more than 18 dB. Some examples are shown in Fig. 8. Compared to method 1, the noise immunity of method 2 was confirmed to be much larger.

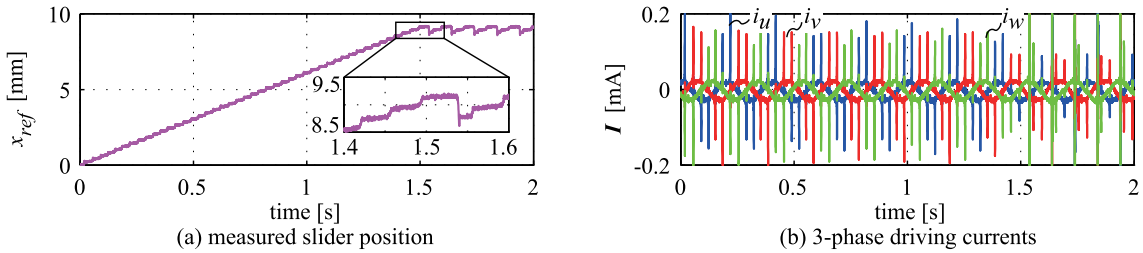
5. Verification using real signal

5.1. Experimental setup

The proposed methods were applied to real driving currents to estimate the slider position. Driving currents and the slider position were measured in an experimental setup. Then, the currents were processed offline using the two methods, whose results were compared with the measured position for verification.

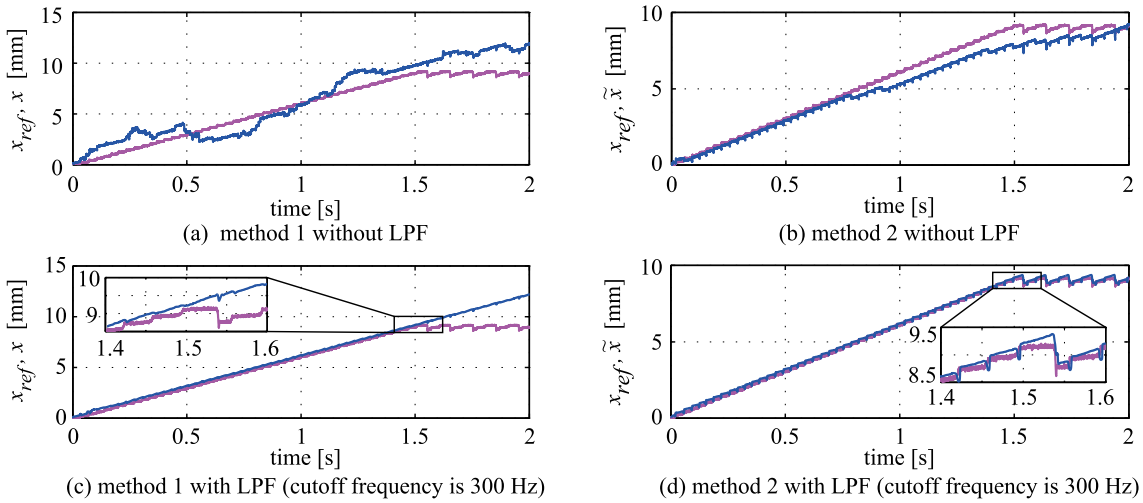
The schematic diagram of the experiment apparatus and its photograph are shown in Fig. 9. The electrode pitch was 0.2 mm, and the slider mass was 25 g, which consisted of the 5-g slider film and the 20-g plastic plate. The effective actuation area, where the slider and the stator electrodes were facing, was 110 cm^2. In the gap between the two films, glass beads with a diameter of 20 µm were sprinkled to reduce friction. A set of two rigid blocks were arranged as a wall such that the slider will step-out when it hits the wall.

The motor was arranged in a dielectric fluid (3M, Flouorinert FC-77) and a 3-phase AC voltage (1500 V, 5 Hz) was applied. The 3-phase voltage was generated by a digital signal processing system (Dspace, ds1104) and then amplified by three high-voltage amplifiers (NF, HVA4321). A laser displacement sensor



(a) measured slider position (b) 3-phase driving currents

Fig. 10. Reference slider position and driving currents.



(a) method 1 without LPF (b) method 2 without LPF

(c) method 1 with LPF (cutoff frequency is 300 Hz) (d) method 2 with LPF (cutoff frequency is 300 Hz)

Fig. 11. Estimated slider position based on two methods.

(Matsushita Electric Works, ANR-1282) measured the slider position for reference. The driving currents were measured using the built-in current meters of the high-voltage amplifiers, which have a full scale of 10 mA and the accuracy of 50 μA, according to the datasheet. The output of the current meters were sampled using the digital signal processing system, with a time step of 0.1 ms.

5.2. *Results*

The measured slider position and corresponding driving currents are shown in Fig. 10. Because the measured driving currents had DC offsets, a HPF (2nd-order Butterworth, cutoff frequency 0.5 Hz) was used to remove the offset. For the method 1, the constant part of $\mathbf{I}_{dq}$ was set as $A = 0.02$ mA. For the method 2, the HPF was a 2nd-order Butterworth filter with a cutoff frequency of 5 Hz.

The estimated positions based on the two methods are shown in Fig. 11(a) and (b) using blue lines. For comparison, the reference slider position measured by the laser sensor is also plotted. Although the method 2 worked better than the method 1, both of the two methods could not correctly trace the slider position. This is due to the noise in the measured currents.

To reduce the noise, the measured currents were band-limited by a low pass filter (LPF, 2nd order, Butterworth) with a cutoff frequency of 300 Hz before processing. The corresponding outputs are shown in Fig. 11(c) and (d). In method 1, the estimation was successful during synchronous motion, but the

method could not correctly track out-of-step motions. On the other hand, method 2 worked well in both synchronous and out-of-step motions.

5.3. Discussion

The results of the two methods show that method 1 does not work well in practical situation as its noise immunity was very low. This is because the denominator in Eq. (17), which is $i_d - A$ is too small when the slider speed is low and thus the calculation of arctan in Eq. (17) can be considerably affected by noise. On the other hand, in method 2, the denominator does not become too small and arctan provides a stable value, which made the method robust against noise. In actual situations, we cannot expect high SNR in current measurement as shown above, as the currents of the electrostatic motor is typically small. Therefore, we can conclude that method 2 is more suitable for actual implementation. However, as discussed earlier, method 2 will not correctly work if a slider is firmly fixed, which should be paid attention to when implementing the method.

The prerequisite for this sensing method is that we are able to measure the driving currents with enough accuracy. As the current amplitude is almost proportional to the slider speed, measurement of the currents becomes easy at higher slider speed. In the stepwise motions discussed in this paper, the slider jumps with a high speed for every step, because of which the sensing method could work robustly. If, however, the slider moves with a smooth linear trajectory, the current measurement will become considerably hard. Smooth trajectory appears when the inertia becomes large, or when the skew motor [22] is used. Displacement estimation for such situations are still left as a challenging problem.

6. Conclusion

This paper demonstrated new methods for displacement estimation of the synchronous electrostatic film motor, using the driving currents. The methods target stepwise slider motions that appear in a motor with the normal (non-skewed) electrode configuration. The methods utilize Clarke and Park transformations to distill the slider position information from the currents. Two different methods were discussed for different noise magnitudes. Numerical simulations and a verification using the real motor currents confirmed that method 1 is sensitive to signal noise and can not work well in reality. Method 2, on the other hand, can work under noise, but its resolution is compromised to plus/minus one electrode pitch.

Compared with the self-sensing method using the driving currents discussed in the previous study, the method in this paper is found superior in that it can correctly detect "out-of-synchronous" motions. Also, compared with other sensing methods proposed for the same electrostatic motor, the method proposed here is characterized by its simplicity of the system structure. Therefore, it would be suitable for relatively low-cost applications that do not require high precision of positioning.

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